

COMPREHENSIVE INDOOR SMOKING BANS AND SMOKING PREVALENCE

Evidence from the BRFSS

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ABSTRACT

In an effort to reduce exposure to secondhand smoke, many governments have enacted smoke-free laws in public spaces. To the extent that smoke-free laws change perceptions, norms, and attitudes about tobacco use, these laws may also reduce the prevalence and intensity of cigarette consumption. Using nationally representative data on over four million individuals spanning 15 years, we study the average effect of comprehensive, state-level indoor smoking bans on smoking prevalence in the United States. Our research design exploits within-state variation in smoking prevalence, as well as the rapid diffusion of comprehensive bans across states between 2002 and 2010. Our results suggest that a comprehensive indoor smoking ban is associated with a 2.35 percent to 3.29 percent average reduction in smoking prevalence, a range that is robust to various specifications. These effects are roughly equivalent to a \$0.95 to \$1.33 increase in the average state's cigarette tax in 2010. Finally, we demonstrate that state-level indoor bans in bars may significantly decrease smoking prevalence in specific subpopulations, particularly young, female, low-income, and binge-drinking individuals.

KEYWORDS: indoor smoking ban, smoking cessation

JEL CLASSIFICATION: I12, I18

I. Introduction

Cigarette smoking remains the principal cause of death in the United States, causing approximately 448,000 deaths each year.¹ Despite the robust, established, and causal relationship between smoking and adverse health outcomes, nearly 18 percent of the adult

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1 For recent evidence on the health effects of smoking, see USDHHS (2004) and USDHHS (2012). Also see the Centers for Disease Control and Prevention, http://www.cdc.gov/tobacco/data_statistics/fact_sheets/fast_facts/.

population in the United States continues to smoke. Beyond the direct health consequences of smoking, secondhand smoke also produces caustic and carcinogenic agents that threaten the health and well-being of nonsmokers (USDHHS 2006). Each year, nearly 50,000 nonsmokers die from the effects of secondhand smoke, adding \$10 billion in extra medical care, illness, and mortality to US national health expenditures.²

Recognizing the negative externalities created by cigarette smoking, policy makers have used a variety of tools to discourage cigarette smoking. While a voluminous literature exists on the price and tax elasticities of cigarette demand, the general consensus is that taxes have been effective at curtailing smoking among light to moderate smokers, and the price elasticity of cigarettes for smokers is small.³ However, if the price elasticity of demand for cigarettes varies across different subgroups of the population (e.g., smoking history, age, gender), raising cigarette taxes may simply have the effect of extracting surplus from certain groups without significantly reducing smoking prevalence. Indeed, for both the overall population and for specific subgroups, policies other than taxation may be important in encouraging smoking cessation.

In addition to cigarette taxes, governments have implemented smoke-free policies that prohibit combustible tobacco smoke in any number of workplaces, restaurants, and/or bars. Naturally, much of the research on smoke-free policies has focused on exposure to secondhand smoke, but the literature cites varying opinions on the impact of these policies. For instance, some studies attribute any observed effect of smoke-free policies to specific subsets of the population: youth (McMullen et al. 2005), women (Albers et al. 2004), and hospitality workers (Anger, Kvansicka, and Siedler 2011). Alternatively, Adda and Cornaglia (2010) find that smoke-free policies can have perverse effects on nonsmokers, especially young children, because an indoor smoking ban increases the probability that a smoker will continue the habit at home rather than in a public place. However, results from the 2012 International Tobacco Control Policy Evaluation Project, a prospective panel study conducted in four countries before and after smoke-free law implementation, show that smoking bans actually tend to prompt smokers to smoke less at home (Mons et al. 2013). Callinan et al. (2010) review the literature in 13 countries and conclude that indoor smoking bans reduce exposure to secondhand smoke in workplaces, restaurants, and public houses.

In addition to secondhand smoke, smoke-free policies may reduce smoking *prevalence* by challenging the perceptions, norms, and attitudes about tobacco use (Fichtenberg and Glantz 2002). Specifically, these policies may de-normalize smoking by changing the attitudes and perceptions held by adults and adolescents. For example, Albers et al. (2004) find that adults and youth predominantly exposed to smoke-free regulations are more likely to perceive smoking as an unacceptable behavior. However, the literature on the impact of indoor smoke-free laws on smoking prevalence is mixed—even in studies that take the endogeneity of smoke-free laws seriously. The public health literature has

2 For more information on the health effects of secondhand smoke, see the American Cancer Society <http://www.cancer.org/cancer/cancercauses/tobaccocancer/secondhandsmoke>.

3 See Callison and Kaestner (2014) for recent evidence on the small price elasticity of demand for cigarettes.

shown that smoke-free laws may promote quitting among smokers, increase the number of quit attempts, decrease smoking intensity, and discourage initiation among youth (McMullen et al. 2005; Farrelly et al. 2013; Elmont et al. 1993). Furthermore, Evans, Farrelly, and Montgomery (1999) find evidence that workplace indoor smoking bans reduce smoking prevalence and daily intensity, even after accounting for the fact that healthier workers may select into firms that enact smoking bans. On the other hand, Tauras (2006) finds little evidence that smoke-free policies reduce the prevalence of adult smoking in the United States, and, more recently, Jones et al. (2015) find no effect of smoke-free laws on smoking consumption in the United Kingdom.

In this paper we study the effect of comprehensive, statewide indoor smoking bans—a ban on smoking in all restaurants, bars, and workplaces—on the prevalence of cigarette smoking. Beginning with Delaware in 2002, there has been steady growth in statewide indoor smoking bans that *completely* prohibit cigarette smoking in private sector work sites, restaurants, and bars. Indeed, by 2010, 46 percent of US states, representing 39 percent of the US population, had enacted a comprehensive ban. Our research design uses repeated cross sections from the Behavioral Risk Factor Surveillance System (BRFSS) from 1996 to 2010 to identify an effect that may plausibly be interpreted as causal. Our pooled data are nationally representative and contain over four million observations, which allows us to identify small effects precisely. Our initial approach uses within-state variation over time in smoking prevalence merged with information on comprehensive smoking bans to identify the effect of bans on prevalence. Using a state-and year-level fixed-effects estimator, we find a negative and significant effect of comprehensive bans on smoking prevalence. At the mean level of smoking prevalence, a comprehensive ban is found to reduce smoking prevalence by 2.35 percent relative to enforcing *no indoor bans* in workplaces, restaurants, or bars. This effect is roughly equivalent to a \$0.95 increase in the average state-level cigarette tax in 2010.⁴ To support our state-level fixed-effects estimator, we show that smoking trends, conditional on observables including cigarette taxes, other state-level smoking policies, and comprehensive indoor smoking bans, are similar in magnitude across states that ever and never enact comprehensive indoor bans.

Our fixed-effects estimator is unbiased if there do not exist any time-varying unobserved variables that are correlated with a state's comprehensive ban status and an individual's propensity to smoke. While we show that conditional trends in smoking prevalence are similar in magnitude between states that ever enact a comprehensive ban and those that do not, our result may still be biased if the *timing* of enactment is nonrandom. To investigate further, we exploit the fact that comprehensive bans rapidly spread across states between 2002 and 2010. Indeed, while no state had enacted a ban prior to 2002, an average of three states per year enacted such bans through 2010. We exploit the spread of comprehensive bans across states over time by estimating a fixed-effects instrumental variables (FEIV) estimator that amounts to a scaled difference-in-differences design. The first difference is between states that are *ever* observed to enact a ban, and the

4 This calculation assumes a price elasticity of demand of -0.035 . See Callison and Kaestner (2014), who find a range between -0.02 and -0.05 .

second difference is before and after 2002—the year of the first state-wide comprehensive ban. Our instruments—interactions between whether a state ever enacts a comprehensive ban and year indicators—scale the difference-in-differences to account for the timing of ban enactment. This identification strategy recognizes both state heterogeneity in smoking decisions—through fixed effects—as well as the possibility of the nonrandom timing of ban enactment—by removing the temporal component of ban enactment. We find a comprehensive ban to be associated with a 3.29 percent reduction in average smoking prevalence at the mean level of smoking. Our overall results suggest that reasonable bounds for the reduction in smoking prevalence associated with a comprehensive ban are 2.35 percent and 3.29 percent.

Our FEIV estimator exploits the fact that, prior to 2002, no state had enacted a state-wide comprehensive indoor smoking ban. The identification assumption is that prior to 2002, trends in smoking were similar by whether a state is ever observed to enact a comprehensive ban. Our FEIV estimator allows us to easily test this assumption. We conduct an overidentification test that includes our pre-2002 instruments in the main smoking equation. In all cases, we fail to reject the null hypothesis of parallel trends. Furthermore, the estimated effect of a comprehensive indoor ban on the propensity to smoke does not change when we allow for differential state smoking trends. As a further robustness check, to rule out the possibility of local government indoor smoking bans confounding our estimated effect, we use the county of residence information in the BRFSS to control for state/county-level fixed effects. Our results from FEIV models that control for county fixed effects are not different, in magnitude or significance from our state-level model. Finally, we demonstrate that either estimator—the fixed-effects or the fixed-effects instrumental variables estimator—reduces the effect size relative to results from an ordinary least squares (OLS) estimator that treats indoor bans as exogenous. Indeed, when we treat a comprehensive ban as exogenous, our OLS estimator suggests a 4.23 percent reduction in smoking prevalence. If states that are more likely to adopt comprehensive indoor smoking bans are different in other, unobserved, ways that systematically vary with smoking prevalence, either because of differential levels of health, attitudes, or historical norms, then OLS will tend to overstate the effect of bans. The difference in magnitude highlights a contribution of our study relative to papers that treat bans as exogenous (McMullen et al. 2005; Farrelly et al. 2013; Elmont et al. 1993).

The BRFSS data allow us to identify subpopulations of smokers who may be particularly sensitive to indoor smoking bans—and particularly insensitive to cigarette taxes. To investigate, we disaggregate our data on state-level indoor smoking bans to their component parts: restaurants, restaurants and bars, and workplaces. Our results suggest that indoor bans in bars are especially influential on smoking behavior. Our results suggest that binge drinkers (those admitting to have drunk at least five drinks in a sitting in the past 30 days), the young (those less than 30 years of age), females, and low-income individuals (those earning less than \$10,000 per year) all smoke significantly less following state-level bans in bars. For example, using our FEIV estimator, we find that while binge drinkers smoke at almost double the overall rate (38.3 percent compared with 21.3 percent), indoor smoking bans in bars reduced binge-drinker smoking by 5.20 percent. This result is consistent with Bitler, Carpenter, and Zavodny (2009), who find evidence

that restrictions specifically on smoking in bars reduced smoking prevalence among bartenders, and Anger, Kvansicka, and Siedler (2011), who identify strong negative effects of indoor bans in Germany for individuals who claim to frequent bars. We also find larger effects when focusing on workplace bans and restricting our sample to just those who are currently employed, although this result is not statistically significant.

In addition to our overall and ban type-specific results, our work contributes to the literature in two other ways. First, our paper uses recent, nationally representative data from the United States. Tauras (2006) studies *partial* US state-level indoor smoking bans in the 1990s that restricted smoking in some but not all public spaces; however, our data suggest that only two states had such policies by 1998 and no states had comprehensive bans.⁵ Importantly, our methodology is similar to Anger, Kvansicka, and Siedler (2011), who use variation in timing of enactment to identify the effect of smoke-free laws on smoking prevalence across German federal states. Those authors find no effect of bans on average smoking rates. Second, we have considerable variation in ban enactment over time. Between 2002 and 2010, an average of three states enacted a comprehensive ban each year. Indeed, comprehensive bans spread across the United States such that by 2010, nearly half of the states were treated. By contrast, Jones et al. (2015) exploit the differential timing of ban enactment between England and Scotland.

This paper proceeds as follows. In Section II, we describe our panel data set of smoking prevalence and indoor smoking ban legislation. We describe our preferred fixed-effects and FEIV estimators in Section III and discuss our estimation results. Section IV covers robustness of our main estimates and extends our model to examine specific subpopulations and specific types of bans. Section V puts these results in context and concludes.

II. Data

Using multiple data sources, all of which are publicly available, we construct a state-level panel data set of smoking legislation from 1996 to 2010 (Robert Wood Johnson Foundation 2013). State legislation regarding indoor smoking bans targeted three types of private establishments: restaurants, restaurants and bars, or workplaces. Table 1 provides population weighted averages by year for each type of state-level indoor ban. Prior to 1998, there did not exist any state-level indoor smoking bans.⁶ Utah and California, representing roughly 12.8 percent of the US population, were the first to implement indoor bans in restaurants and restaurants and bars, respectively, in 1998. By 2010, 71.4 percent of the country lived in a state with at least one type of indoor ban, and 39 percent of individuals lived in a state with all three types of indoor bans.

Naturally, our three indoor ban variables are highly correlated. Indeed, of the 23 states that are *ever* observed to enact all three types of bans, 18 enacted all three at once. Therefore, we categorize policies based on their coverage; we define a comprehensive law as covering all workplaces, bars, and restaurants and a partial law as covering

5 In our empirical analysis, we control for partial indoor smoking bans. See Section III for more details.

6 We discuss the implication of local municipal bans in Section IV.

TABLE 1. Population weight state-level characteristics

Year	Indoor smoking ban type				Other characteristics			
	Restaurants (1)	Restaurants and bars (2)	Workplaces (3)	Partial ban only (4)	Comprehensive ban (5)	Real Cigarette tax \$ (6)	Unemployment rate (7)	Poverty rate (8)
1996	0.000	0.000	0.000	0.000	0.000	0.390 [0.035]	5.426	13.671
1997	0.000	0.000	0.000	0.000	0.000	0.402 [0.035]	4.970	13.219
1998	0.128	0.120	0.000	0.128	0.000	0.410 [0.037]	4.530	12.700
1999	0.128	0.120	0.000	0.128	0.000	0.469 [0.064]	4.240	11.788
2000	0.129	0.121	0.000	0.129	0.000	0.492 [0.073]	4.021	11.241
2001	0.129	0.121	0.000	0.129	0.000	0.484 [0.070]	4.768	11.678
2002	0.132	0.124	0.005	0.132	0.003	0.675 [0.086]	5.794	12.114
2003	0.270	0.191	0.130	0.203	0.069	0.706 [0.087]	5.996	12.455
2004	0.301	0.229	0.153	0.213	0.090	0.742 [0.096]	5.528	12.739
2005	0.331	0.254	0.183	0.222	0.114	0.801 [0.092]	5.104	12.606
2006	0.427	0.341	0.272	0.246	0.186	0.796 [0.093]	4.632	12.278
2007	0.489	0.388	0.323	0.272	0.222	0.871 [0.086]	4.631	12.472
2008	0.560	0.458	0.435	0.314	0.292	0.950 [0.120]	5.780	13.215
2009	0.577	0.488	0.459	0.295	0.328	1.055 [0.114]	9.263	14.324
2010	0.670	0.581	0.518	0.324	0.390	1.152 [0.175]	9.667	15.127

Notes: Statistics are calculated from all 50 states over 15 years, n = 750. All statistics are weighted by state population. Each column represents the proportion of the United States population that live in a state where the respective indoor ban is in effect. A comprehensive indoor smoking ban covers all workplaces, bars, and restaurants. A partial ban is defined as at least one type of ban in effect, but not all three. Real cigarette taxes represent state-level taxes per pack represented in 2000 dollars. Standard deviations are in brackets.

one or two of the preceding establishments. However, our indicator for a partial ban is not cumulative. For the five states that transition from a partial ban to a comprehensive ban, the partial ban indicator turns off when the third type of ban is enacted. Columns 4 and 5 of Table 1 report the population weighted proportions for partial and comprehensive bans. In 2002, Delaware became the first state to enact a comprehensive ban, and, on average, three states enacted comprehensive bans in each following year.

We also collect state-level cigarette tax data from the Centers for Disease Control and Prevention's tracking system. The data are expressed in real dollars per pack (CDC 2014). Taxes are expressed in year 2000 dollars using the Bureau of Labor Statistics Consumer Price Index for all US items and are weighted by population (US Department of Labor 2015). Because we ultimately identify our models from within-state variation in smoking prevalence, we focus only on state-level cigarette taxes. Finally, to control for state economic characteristics that vary over time, we collect state poverty and unemployment data from the US Census and the Bureau of Labor Statistics, respectively (Bureau of Labor Statistics 2014).

Next, we collect individual-level repeated cross-sections from the Behavioral Risk Factor Surveillance System (BRFSS), a yearly phone survey that monitors health behaviors for each state. These data are nationally representative, and our pooled data set includes 4,049,706 observations. We restrict our sample to adults aged 18 or over who have nonmissing information on current smoking behavior. The data include information on an individual's demographic (e.g., age, gender, ethnicity), socioeconomic (e.g., education, income, marital status), and health (e.g., pregnancy, smoking, and alcohol consumption) characteristics. Table 2 provides weighted estimates of population proportions for each characteristic using the BRFSS weights. Rather than drop individuals with missing values (aside from those with missing smoking information), we generate missing value dummy variables for each relevant characteristic. 51.6 percent of our sample is female and 9.6 percent is African American. 30.4 percent of our sample has at least graduated from college. Binge drinking is defined as consuming more than five drinks in one sitting in the past 30 days.

Figure 1 shows the mean (weighted) smoking prevalence by year, which declines by 6 percentage points from 1996 to 2010. Our smoking data are consistent with the general slowing in the decline in cigarette smoking. Between 1965 and 1995, smoking declined by nearly 41.27 percent; our data (consistent with CDC estimates) suggest only a 26.8 percent decline between 1996 and 2010 (CDC 2016).

III. Econometric Methods and Results

Economic theory provides a guide with respect to modeling smoking behavior.⁷ In addition to standard constraints with respect to income and prices, a central trade-off in the

7 In addition to the seminal rational addiction theory of Becker and Murphy (1988), other economic models of smoking capture regret and learning, Orphanides and Zervos (1995); time-inconsistent preferences, Gruber and Koszegi (2001); cue-triggered mistakes, Bernheim and Rangel (2004); compulsive consumption, temptation, and self-control, Gul and Pesendorfer (2001, 2007), and learning Darden (forthcoming).

TABLE 2. Weighted proportions of explanatory variables

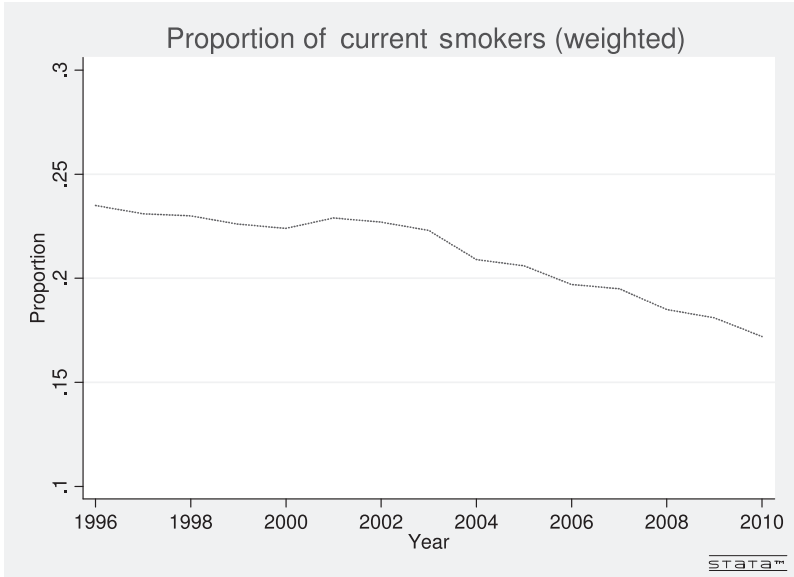
Age		Race	
<30	0.209	African American	0.096
[30,40)	0.201	Other non white	0.175
[40,50)	0.200	Missing	0.007
[50,60)	0.160	Income	
[60,70)	0.112	<\$10k	0.048
[70,80)	0.082	[\$10k, \$15k)	0.048
≥80	0.037	[\$15k, \$20k)	0.068
Female	0.516	[\$20k, \$25k)	0.084
Married	0.595	[\$25k, \$35k)	0.120
Missing	0.000	[\$35k, \$50k)	0.147
Education		[\$50k, \$75k)	0.147
<High school	0.123	>75k	0.203
High school	0.303	Missing	0.134
Some college	0.267	Binge drinker	0.127
College or more	0.304	Missing	0.557
Missing	0.003	Pregnant	0.012
Currently working	0.613	Missing	0.269
Missing	0.005		

Notes: Proportions are from the BRFSS and are weighted using BRFSS weights. Rather than drop observations with missing values, we create dummy variables for all variables with missing values. The total sample size from 15 repeated cross sections is $n = 4,049,706$.

decision to smoke cigarettes is an intertemporal one: the enjoyment of current smoking today as compared with the potential health (and utility) consequences tomorrow. The degree to which (and consistency with which) an individual discounts the future plays a fundamental role in the smoking decision, as well as the extent to which an individual is fully informed of the associated health risks. Furthermore, addiction—classically defined in Becker and Murphy (1988) as an increase in the marginal utility of smoking in past smoking that is sufficiently large—changes the “enjoyment of current smoking.” Our paper is not geared towards understanding the structural mechanisms of information, learning, time-preference, and addiction—mainly because of data limitations—but we argue that careful econometric evidence of shifts in smoking behavior caused by public policy (e.g., indoor smoking bans) can inform future structural approaches to addiction more broadly.⁸

8 See the voluminous literature in the spirit of capturing the determinants of smoking cessation by Sloan and colleagues (Khwaja, Sloan, and Chung (2006), Arcidiacono, Sieg, and Sloan (2007), and Sloan, Smith, and Taylor (2003)) and Viscusi and colleagues (Viscusi (1990) and Viscusi and Hakes (2008)).

FIGURE 1. Smoking proportions by year of US adults



Notes: Figure displays the proportion of the US adult population described as current smokers by year. Proportions are from authors' calculations from the Behavioral Risk Factor Surveillance System (BRFSS). All proportions are weighted using BRFSS weights.

In that spirit, our initial empirical model of smoking behavior focuses on the binary, static decision by individual i , living in state s , to smoke or not in year t , S_{ist} . Equation 1 presents our initial model:

$$S_{ist} = \alpha_s + \beta_1 1[\text{Complete Ban}_{st}] + \beta_2 1[\text{Partial Ban}_{st}] + X_{ist}\delta + \gamma_t 1[\text{Year} = t] + \epsilon_{ist} \quad (1).$$

Here, the smoking decision is a function of state-level fixed effects, α_s ; binary variables for whether a complete or partial state-level ban is in effect in state s at time t ; individual- and state-level controls, X_{ist} ; year dummy variables, $1[\text{Year} = t]$; and an i.i.d. error term, ϵ_{ist} . Year effects capture the secular downward trend in smoking over time. β_1 is our coefficient of interest, which is identified from within-state variation in smoking prevalence over time. Because we control for $1[\text{Partial Ban}_t]$, β_1 can be interpreted as capturing the effect of a state enacting a comprehensive ban relative to the absence of any indoor bans.

The state fixed effects control for time-invariant state-level unobserved variables that may be correlated with both an individual's decision to smoke and whether a state has a comprehensive ban. Table 3 presents evidence of the importance of state fixed effects. The table presents the mean of key variables by whether the state is *ever* observed to have a comprehensive ban between 1996 and 2010. States that ever have a comprehensive ban have, on average, significantly lower rates of smoking; higher state-level cigarette

TABLE 3. Weighted statistics by ever enacting a comprehensive ban

Ever enacts comprehensive ban	Smoking prevalence	Cigarette tax	Unemployment rate	Poverty rate
No	22.874	0.438	5.413	13.097
Yes	20.211	0.935	5.063	11.005
<i>p</i> -value	0.000	0.000	0.010	0.000

Note: Table reports means by whether a state is ever observed to enact a comprehensive indoor smoking ban between 1996 and 2010.

taxes; and lower rates of unemployment and poverty. If states that enact a comprehensive ban have populations that both smoke less and also demand that their state governments enact comprehensive bans—driven perhaps by cultural, historical, or other unmeasured factors—then failing to include state-specific effects will tend to overstate the effect of comprehensive bans on smoking prevalence.

The first two columns of Table 4 present results from an ordinary least squares regression of equation 1 without and with state fixed effects, respectively.⁹ In column 1, both a comprehensive and a partial ban significantly reduce the probability of smoking. Smoking is decreasing in age, education, and income. Individuals who are married, pregnant, or African American are all less likely to smoke. Binge drinking is positively associated with smoking.

Column 2 adds state-level fixed effects to the model in column 1. While point estimates on our control variables do not change significantly between columns 1 and 2, the effect of comprehensive and partial indoor smoking law drops dramatically. Indeed, at the mean smoking proportion of 0.213, the effect of a comprehensive ban drops from a 4.23 percent reduction to a 2.35 percent reduction. The effect of a partial ban becomes statistically indistinguishable from zero. Including state fixed effects also reduces the effects of the cigarette tax rate and the rates of unemployment and poverty.

Our parameter estimates in column 2 of Table 4 that control for state fixed effects can be interpreted as causal if there do not exist any time-varying, unobserved factors that are correlated with both a state having a comprehensive indoor smoking ban in year t and the decision to smoke cigarettes. To investigate this assumption, equation 2 relaxes the assumption of common year effects in smoking prevalence by interacting year dummy variables with whether a state is ever observed to have a comprehensive ban between 1996 and 2010.

$$\begin{aligned} S_{ist} = & \alpha_s + \beta_1 1[\text{Complete Ban}_{st}] + \beta_2 1[\text{Partial Ban}_{st}] + X_{ist} \delta + \gamma_t 1[\text{Year} = t] \\ & + \omega_t 1[\text{Year} = t] \text{Ever}_{st} + \epsilon_{ist} \end{aligned} \tag{2}.$$

9 All estimates come from linear probability models. We arrive at qualitatively similar results using logit models, but we prefer the linear probability model when using fixed effects. All regressions are weighted using the BRFSS weights, and all standard errors are clustered at the state/strata level.

TABLE 4. Estimates of the effect of a comprehensive smoking ban on mean smoking prevalence

Mean (weighted) smoking proportion = 0.213					
	[1]	[2]	[3]	[4]	[5]
Comprehensive ban	-0.009 ^a (0.001)	-0.005 ^a (0.002)	-0.004 ^b (0.002)	-0.007 ^a (0.002)	-0.008 ^a (0.003)
Partial ban	-0.034 ^a (0.001)	0.003 (0.002)	0.002 (0.002)	0.003 (0.002)	0.003 (0.002)
Age					
[30,40)	0.039 ^a (0.001)	0.039 ^a (0.001)	0.039 ^a (0.001)	0.039 ^a (0.001)	0.039 ^a (0.001)
[40,50)	0.055 ^a (0.002)	0.055 ^a (0.002)	0.055 ^a (0.002)	0.055 ^a (0.002)	0.055 ^a (0.002)
[50,60)	0.031 ^a (0.002)	0.031 ^a (0.002)	0.031 ^a (0.002)	0.031 ^a (0.002)	0.031 ^a (0.002)
[60,70)	-0.049 ^a (0.002)	-0.049 ^a (0.002)	-0.049 ^a (0.002)	-0.049 ^a (0.002)	-0.049 ^a (0.002)
[70,80)	-0.138 ^a (0.002)	-0.139 ^a (0.002)	-0.139 ^a (0.002)	-0.139 ^a (0.002)	-0.139 ^a (0.002)
≥80	-0.201 ^a (0.002)	-0.201 ^a (0.002)	-0.201 ^a (0.002)	-0.201 ^a (0.002)	-0.201 ^a (0.002)
Female	-0.021 (0.001)	-0.021 ^a (0.001)	-0.021 ^a (0.001)	-0.021 ^a (0.001)	-0.021 ^a (0.001)
Education					
High school	-0.045 ^a (0.002)	-0.046 ^a (0.002)	-0.046 ^a (0.002)	-0.046 ^a (0.002)	-0.046 ^a (0.002)
Some college	-0.095 ^a (0.002)	-0.094 ^a (0.002)	-0.094 ^a (0.002)	-0.094 ^a (0.002)	-0.094 ^a (0.002)
College or more	-0.182 ^a (0.002)	-0.182 ^a (0.002)	-0.182 ^a (0.002)	-0.182 ^a (0.002)	-0.182 ^a (0.002)
Missing	-0.079 ^a (0.008)	-0.071 ^a (0.008)	-0.072 ^a (0.008)	-0.071 ^a (0.008)	-0.071 ^a (0.008)
Race					
African American	-0.054 ^a (0.001)	-0.057 ^a (0.001)	-0.057 ^a (0.001)	-0.057 ^a (0.001)	-0.057 (0.001)
Other non white	-0.078 ^a (0.001)	-0.071 ^a (0.001)	-0.071 ^a (0.001)	-0.071 ^a (0.001)	-0.071 (0.001)

TABLE 4. Continued

	Mean [weighted] smoking proportion = 0.213				
	[1]	[2]	[3]	[4]	[5]
Missing	-0.014 ^a (0.004)	-0.014 ^a (0.004)	-0.014 ^a (0.004)	-0.014 ^a (0.004)	-0.014 (0.004)
Income					
[\$10k, \$15k]	-0.001 (0.003)	-0.002 (0.003)	-0.002 (0.003)	-0.002 (0.003)	-0.002 (0.003)
[\$15k, \$20k]	0.006 ^b (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)	0.003 (0.003)
[\$20k, \$25k]	-0.002 (0.003)	-0.005 (0.003)	-0.005 (0.003)	-0.005 (0.003)	-0.005 (0.003)
[\$25k, \$35k]	-0.019 ^a (0.003)	-0.021 ^a (0.003)	-0.021 ^a (0.003)	-0.021 ^a (0.003)	-0.021 ^a (0.003)
[35k, 50k]	-0.039 ^a (0.003)	-0.041 ^a (0.003)	-0.041 ^a (0.003)	-0.041 ^a (0.003)	-0.041 ^a (0.003)
[\$50k, \$75k]	-0.069 ^a (0.003)	-0.071 ^a (0.003)	-0.071 ^a (0.003)	-0.071 ^a (0.003)	-0.071 ^a (0.003)
>\$75k	-0.105 ^a (0.003)	-0.107 ^a (0.003)	-0.107 ^a (0.003)	-0.107 ^a (0.003)	-0.107 ^a (0.003)
Missing	-0.064 ^a (0.002)	-0.068 ^a (0.002)	-0.068 ^a (0.002)	-0.068 ^a (0.002)	-0.068 ^a (0.002)
Binge drinker	0.150 ^a (0.002)	0.150 ^a (0.002)	0.150 ^a (0.002)	0.150 ^a (0.002)	0.150 ^a (0.002)
Missing	-0.030 ^a (0.001)	-0.033 ^a (0.001)	-0.033 ^a (0.001)	-0.033 ^a (0.001)	-0.033 ^a (0.001)
Pregnant	-0.068 ^a (0.003)	-0.066 ^a (0.003)	-0.066 ^a (0.003)	-0.066 ^a (0.003)	-0.066 ^a (0.003)
Missing	-0.013 ^a (0.001)	-0.012 ^a (0.001)	-0.012 ^a (0.001)	-0.012 ^a (0.001)	-0.012 ^a (0.001)
Working	-0.007 ^a (0.001)	-0.007 ^a (0.001)	-0.007 ^a (0.001)	-0.007 ^a (0.001)	-0.007 ^a (0.001)
Missing	0.004 (0.007)	-0.001 (0.007)	0.000 (0.007)	-0.001 (0.007)	-0.001 (0.007)
Married	-0.076 ^a (0.001)	-0.076 ^a (0.001)	-0.076 ^a (0.001)	-0.076 ^a (0.001)	-0.076 ^a (0.001)
Missing	-0.001 (0.028)	0.002 (0.028)	0.002 (0.028)	0.002 (0.028)	0.002 (0.028)

TABLE 4. Continued

	Mean [weighted] smoking proportion = 0.213				
	[1]	[2]	[3]	[4]	[5]
Real cig tax	-0.013 ^a (0.001)	-0.005 ^a (0.001)	-0.004 ^b (0.002)	-0.004 ^a (0.002)	-0.004 ^a (0.002)
Unemployment rate	-0.002 ^a (0.000)	-0.003 ^a (0.001)	-0.003 ^a (0.001)	-0.003 ^a (0.001)	-0.003 ^a (0.001)
Poverty rate	0.000 ^b (0.000)	0.001 ^b (0.000)	0.001 ^b (0.000)	0.001 ^b (0.000)	0.001 ^b (0.000)
Constant	0.469 ^a (0.004)	0.481 ^a (0.006)	0.477 ^a (0.007)	0.481 ^a (0.006)	0.481 ^a (0.006)
Year FE	Yes	Yes	Yes	Yes	Yes
State FE	No	Yes	Yes	Yes	Yes
<i>p</i> -value			0.020		0.201

Notes: All regressions include year dummy variables. Standard errors are clustered at the combined BRFSS stratum state level. Column 1 reports OLS results. Column 2 reports results when state fixed effects are included. Column 3 includes year and state fixed effects, as well as interactions between year dummy variables and whether a state ever enacts a comprehensive ban. The *p*-value in column 3 is of the null hypothesis that smoking prevalence trends in states that ever enact a comprehensive ban are the same as those that never enact a ban. Column 4 reports results from our preferred FEIV estimator. Column 5 includes year/ever interactions from 1996 to 2001—prior to any comprehensive bans in the main equation. The *p*-value in column 5 is of the null hypothesis that year $\times$ ever interactions between 1996 and 2001 are zero in the main equation. The *F*-statistic in the first stage of column 4 of the excluded variables is 129.71. $n = 4,049,706$. ^a $p < 0.01$, ^b $p < 0.05$.

Intuitively, evidence of statistically similar trends in smoking prevalence conditional upon a comprehensive ban, our X variable controls, and state-level fixed effects would lend evidence to the identification assumption of no time-varying unobserved factors—the presence of which would tend to cause differential trends even when conditioning on a comprehensive ban. Column 3 of Table 4 presents results from the estimation of equation 2. The p -value on the null hypothesis of common year effects is 0.02, clearly rejecting the null hypothesis of common trends at the 95 percent significance level.¹⁰ Importantly, however, the point estimate of 0.004 on a comprehensive indoor ban is still statistically significant at the 95 percent level and not significantly different from the point estimate with the restriction of a common time trend. Furthermore, Meer and West (2013) show that if a treatment effect appears in changes rather than levels, then separate trends will bias estimates towards zero, which may explain the small decline in our point estimate. Finally, given our sample size of over four million observations, it is not surprising that we are able to detect small deviations from the trend in states that never enact a comprehensive ban.

Even if the common trends assumption is valid in this setting, the fixed-effects model in equation 1 may still be biased if there exist shocks to individual smoking decisions that coincide with the enactment of a comprehensive ban. That is, we might be concerned about biased estimates if the timing of the adoption of a comprehensive ban was not random. For example, if a private campaign to reduce smoking simultaneously influenced legislators to enact a comprehensive ban and was successful in reducing smoking prevalence, then our fixed-effects estimator would be unable to distinguish the two effects.

To proceed, we follow similar methodology as in Kaestner, Darden, and Lakdawalla (2014) and exploit the fact that our data span a period in which no comprehensive, statewide smoking bans were in effect (1996–2001), followed by a period of diffusion of these policies across the United States (2002–2010). Indeed, by 2010, nearly half of the states had enacted a complete smoking ban, with an average of two to three states enacting a comprehensive ban each year. We exploit this temporal variation in adoption to identify the effect of comprehensive bans on smoking prevalence. Intuitively, we estimate a scaled difference-in-differences estimator in which the first difference is before and after 2002, and the second difference is between states that ever enact a comprehensive ban and those that never enact a comprehensive ban.¹¹ Because not all states that ever enact a complete ban do so in 2002, we scale the effect by year interactions with the dummy variable for ever enacting a comprehensive ban. In practice, this scaling removes the nonrandom timing component of a particular state's ban.

The empirical model that implements the scaled difference-in-differences specification is a fixed-effects instrumental variables (FEIV) estimator in which we instrument for a comprehensive ban with interactions between year dummy variables and a binary

10 Specifically, we conduct an F -test that $\omega_{1997}, \dots, \omega_{2010}$ are jointly zero.

11 Again, we control for partial ban states. Our results are not sensitive to excluding the small number of states that progress from a partial to a comprehensive ban. Results are available upon request.

variable for ever enacting a comprehensive ban. Algebraically, the model is as follows:

$$S_{ist} = \alpha_s + \beta_1 1[\text{Complete Ban}_{st}] + \beta_2 1[\text{Partial Ban}_t] + X_{ist}\delta + \gamma_t 1[\text{Year} = t] + \epsilon_{ist} \quad (3).$$

$$\text{Complete Ban}_{st} = \phi_s + X_{ist}\pi + \psi_t 1[\text{Year} = t] + \lambda_t 1[\text{Year} = t] \text{Ever}_s + v_{st} \quad (4).$$

In the first-stage model in equation 4, we estimate a linear probability model of state s having a comprehensive ban in year t as a function of $1[\text{Year} = t] \times \text{ever}_s$ interactions. These instruments strongly predict whether a state has a comprehensive ban, but, conditional upon the predicted probability of a comprehensive ban, we exclude the interactions from the main estimating equation in equation 3.¹² Both stages contain controls for state fixed effects, exogenous individual- and state-level characteristics, and a partial, state-level indoor ban. Because states that ever adopt comprehensive bans do so gradually after 2002, the FEIV estimator scales the difference-in-differences estimate by the *predicted* probability of a ban in year t .

Column 4 of Table 4 presents results from our FEIV estimator. We find that a comprehensive indoor ban is associated with a 3.29 percent decline in the probability of smoking that is statistically significant at all standard levels of significance. Point estimates on most of our control variables are consistent with estimates in columns 1 through 3—smoking is negatively related to age, education, income, and being nonwhite, employed, married, and pregnant. As in columns 2 and 3, the effect of a partial ban is not statistically different from zero.

As with standard difference-in-differences estimators, the identifying assumption of our FEIV estimator is that trends in smoking prevalence across states that will ever enact a comprehensive ban and states that will never do so are statistically similar prior to the implementation of the first comprehensive ban. The nice aspect of the FEIV model is that testing for common “pre” trends amounts to an overidentification test of (some of) our instruments. Recall that we have 14 instruments: interactions between year dummies and the dummy variable for a state ever enacting a comprehensive ban. Because no state had a comprehensive ban prior to 2002, we can include our interaction instruments prior to 2002 into the main equation to test for common trends. The overidentification test is an F -test that the parameters on these interaction variables are jointly zero. Column 5 of Table 4 presents results from this model as well as the p -value on the overidentification test. Our estimate of the parameter on comprehensive ban increases by only 0.001—less than 0.5 percent of the mean smoking prevalence—and, with over four million observations, we fail to reject the null hypothesis that trends are similar.¹³

12 The F -statistic on the null hypothesis that the 14 instruments are jointly zero in the first-stage regression is 63.56. The 5 percent critical value for the Stock and Yogo (2005) null hypothesis of weak instruments is 48.01, well below our clustered F -statistic of 63.56.

13 The p -value is 0.201. Point estimates of year interactions in our main equation are between 0.0009 and 0.007. The results, along with results from our first stage, are available upon request.

IV. Robustness and Extensions

Our effect of interest is interpreted as the effect of a state simultaneously enacting indoor smoking bans in restaurants, bars, and workplaces relative to the absence of such bans. Based on evidence in the previous section, we argue that a reasonable range for this effect is a reduction in smoking of between 2.35 percent and 3.29 percent of the mean of smoking. Despite passing overidentification and weak instrument tests, we may still be concerned that local government smoking policy that preempts state-level indoor smoking bans may bias our results. Luckily, contained within the publicly available BRFSS data are indicators that reveal an individual's county of residence. To investigate further, we estimate versions of equations 1 and 3 in which we replace state-level fixed effects with county-level fixed effects. Unfortunately, county information is missing in roughly 16 percent of observations, so our results from this specification are not directly comparable to the results in Table 4. Encouragingly, however, the county-level fixed-effects estimator yields exactly the same point estimate, 0.005, as our state-level fixed-effects estimator. That result is also statistically significant at the 95 percent level. Results from the county-level FEIV estimator are also nearly identical to results in Table 4, column 4. These results are available upon request.

Of particular interest to smoking policy makers are the effects of indoor bans on specific subgroups of the population. Indeed, if indoor bans convince smokers to quit who are otherwise unresponsive to increases in cigarette taxes, then implementing indoor bans in states that have not already done so may be an important tool for policy makers. To investigate further, we examine the effects of *specific* types of indoor bans on *specific* types of individuals. We augment our FEIV model with indicator variables for each type of ban (restaurants, restaurants and bars, and workplaces), and we instrument for each with interactions between whether a state ever enacts *that specific type of ban* and year indicators. Our resulting estimator has three endogenous variables and 42 instruments. As shown in Table 1, the “pre-period” for restaurants and restaurants and bars is 1996–1998, whereas the pre-period for workplaces is 1996–2001.

Table 5 presents results from our FEIV estimator for our overall sample and for specific subgroups of the population. Overall, we find that bars are the main driver in the observed negative relationship between indoor smoking bans and smoking prevalence. Recall that our restaurant and bars indicator is cumulative with the restaurant only indicator, identification of which comes from states that only institute restaurant bans. Summing the two point estimates, we find that, for our overall sample, indoor bans in bars reduce smoking prevalence by 5.63 percent at the mean of smoking.¹⁴

The rest of Table 5 presents results by specific subgroup. First, note that smoking prevalence is dramatically decreasing in age, income, and education, while binge drinkers have the highest rate of smoking (38.3 percent). While Callison and Kaestner (2014) find no evidence that the elasticity of demand for cigarettes varies by age, we find that smoking declines dramatically (10.64 percent) in young individuals (age less than 30) following

14 This estimate is not directly comparable to our overall estimates in Section III because the first state-level restaurant bans occurred in 1998, whereas the first state-level comprehensive bans occurred in 2002.

TABLE 5. Specific indoor bans on smoking prevalence

	Age					
	Overall	<30	[30,40)	[40,50)	[50,60)	[60,70)
Restaurants	0.000 (0.004)	0.016 (0.012)	0.006 (0.010)	−0.008 (0.009)	−0.013 (0.010)	−0.006 (0.009)
Restaurants and bars	−0.012 ^a (0.003)	−0.042 ^a (0.010)	0.001 (0.008)	0.001 (0.008)	−0.016 ^b (0.008)	0.001 (0.007)
Workplaces	0.002 (0.003)	−0.017 (0.009)	−0.012 (0.007)	0.016 ^b (0.007)	0.015 ^b (0.007)	0.006 (0.007)
Mean	0.213	0.263	0.233	0.242	0.212	0.156
<i>n</i>	4,049,706	454,692	638,575	775,710	793,221	653,826
	Age			Income		
	[70,80)	>80	<\$10k	\$10k−\$15k	\$15k−\$20k	\$20k−\$25k
Restaurants	−0.021 ^a (0.008)	−0.011 (0.009)	0.005 (0.022)	−0.017 (0.020)	−0.009 (0.018)	−0.017 (0.015)
Restaurants and bars	0.000 (0.006)	0.011 (0.007)	−0.057 ^a (0.018)	−0.021 (0.017)	−0.006 (0.015)	−0.006 (0.012)
Workplaces	0.020 ^a (0.006)	0.002 (0.007)	0.014 (0.016)	0.028 (0.015)	0.012 (0.014)	0.023 ^b (0.012)
Mean	0.091	0.043	0.292	0.275	0.283	0.272
<i>n</i>	480,384	253,298	197,439	217,281	287,794	359,009
	Income					
	\$25k−\$35k	\$35k−\$50k	\$50k−\$75k	≥\$75	Workers	Binge drinkers
Restaurants	−0.007 (0.012)	−0.016 (0.011)	0.000 (0.010)	−0.007 (0.009)	0.007 (0.006)	0.007 (0.015)
Restaurants and bars	0.002 (0.011)	0.001 (0.009)	−0.009 (0.008)	0.003 (0.007)	−0.008 (0.005)	−0.026 ^b (0.013)
Workplaces	0.001 (0.010)	0.007 (0.008)	0.007 (0.007)	−0.004 (0.006)	−0.008 (0.004)	0.001 (0.011)
Mean	0.252	0.226	0.185	0.128	0.223	0.383
<i>n</i>	493,430	600,840	581,737	763,577	2,274,368	427,349
	Education					
	Women	<High school	High school	Some college	College or more	
Restaurants	−0.003 (0.005)	0.003 (0.015)	−0.014 (0.008)	0.009 (0.008)	−0.004 (0.006)	
Restaurants and bars	−0.013 ^a (0.004)	−0.020 (0.012)	−0.013 (0.007)	−0.020 ^a (0.006)	0.008 (0.005)	

TABLE 5. *Continued*

		Education			
	Women	<High school	High school	Some college	College or more
Workplaces	0.003 (0.004)	−0.003 (0.012)	0.011 (0.006)	−0.001 (0.006)	−0.002 (0.004)
Mean	0.189	0.295	0.269	0.221	0.109
<i>n</i>	2,473,027	427,340	1,255,341	1,088,545	1,269,010

Notes: All estimates come from our FEIV estimator. Standard errors are clustered at the combined BRFSS stratum state level. Each set of estimates comes from separate regressions. The *p*-value on the overidentification test of common trends prior to each type of ban is 0.643. ^a*p* < 0.01; ^b*p* < 0.05.

a state-level indoor smoking ban at bars. Not surprisingly, we also find that indoor bans in bars significantly reduce smoking among binge drinkers (roughly 5.20 percent). We also find indoor bans in bars to significantly reduce smoking in female, low-income (less than \$10,000 per year), and “some college” individuals. Smoking in employed individuals declines after workplace indoor bans, but not significantly. We find that workplace bans *increase* smoking in certain age and income groups, but we caution that multicollinearity may be an issue with workplace bans as there exists less variation in workplace bans across states.¹⁵

V. Discussion and Conclusion

This study utilizes nationally representative data from the Behavioral Risk Factor Surveillance System to estimate the effect of state-level comprehensive indoor smoking bans on the prevalence of smoking. Results demonstrate that comprehensive smoke-free policies are responsible for a 2.35 percent (Table 4, column 2) to 3.29 percent reduction (column 4) in smoking prevalence at the mean level of smoking prevalence. In addition, subgroup analyses reveal a significant protective effect of comprehensive laws on groups often targeted by tobacco-control programs—the young, low-income, and binge drinkers. Relative to a 4.35 percent reduction from an estimator that treats smoking bans as exogenous, our results suggest that studies that do not properly control for the endogeneity of smoking-related policies overrepresent the impact of these laws.

Previous research investigating the impact of state-level smoke-free policies on smoking prevalence has been inconsistent. The evidence on reduced intensity is well established, but the link with prevalence is harder to corroborate. In addition, smoke-free laws have been shown to have less impact than current levels of tobacco taxes at reducing prevalence or intensity (Tauras 2006). The results from our research suggests (1) a strong negative

15 Models in which we only instrument for workplace bans while controlling for other types of bans consistently show negative associations with smoking.

association between comprehensive smoke-free policies and smoking and (2) an impact of comprehensive smoke-free policies equal to a \$0.95 to \$1.33 increase in tobacco taxes. Furthermore, we find that indoor bans in bars may be the most effective at reducing smoking. We find important declines in smoking after indoor bar bans in young, some college education, female, low-income, and binge-drinking subpopulations.

Our study has two main limitations. First, we note the temporal aspects of our empirical model. The legislative climate regarding smoke-free policies during our time frame allows the unique opportunity to evaluate the impact of comprehensive laws before and after their implementation. Thus, our estimates must be interpreted as relatively short-term effects. Second, for data reasons, we are unable to examine the extent to which smokers change the intensity with which they smoke. Despite these limitations, this research has implications for tobacco-control policy, and our results serve as an appropriate foundation for future studies.

Finally, while our paper focuses on comprehensive indoor smoking bans, partial smoke-free laws have also become more widespread at the state level. We argue that partial bans are less effective than comprehensive laws for two potential reasons. First, smokers can easily avoid the inconvenience of a partial law. If one is not permitted to smoke in restaurants but permitted to smoke in bars, then one could merely change patronage patterns and choose bars over restaurants (especially where bars find innovative ways to circumvent the partial law by offering off-site food). But if the law is comprehensive, changing patronage patterns serves no purpose, and there is no way to circumvent the law. Second, partial laws do not adequately marginalize smoking. With smoking ubiquitous in entertainment venues, individuals that frequent these venues learn to accept smoking as a norm in their location, whereas limiting smoking everywhere normalizes nonsmoking behavior.

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